Lab 2: DeepConvLSTM Ablation Study

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10/27/2026

Part A: Is the LSTM what helps?

Replacing the two LSTM layers with two dense layers lowered test weighted F1 to 0.869 and macro F1 to 0.457. That is 0.020 and 0.119 below the reference, even though the dense model has almost twice as many parameters. The reference model reached a test weighted F1 of 0.889 and a macro F1 of 0.576, and across the three seeds these varied by about 0.002 and 0.02, so we treat differences of that size as noise. Replacing the LSTM cells with GRU cells gave 0.893 weighted F1 and 0.604 macro F1 with 25% fewer parameters. That is slightly higher than the reference, but not significantly so with three seeds per model. Therefore, recurrent layers improve recognition, and the benefit goes mostly to the gestures, since macro F1 drops much more than weighted F1 without them. However, the benefit does not depend on LSTM cells specifically, because a GRU works at least as well.

Part B: Do the convolutions matter?

With no convolutional layers, the LSTM reads the 113 raw channels directly. This model reached only 0.805 weighted F1 and 0.220 macro F1, barely above always predicting Null (0.757 weighted F1), and 8 of the 17 gestures were never recognized correctly. With one conv layer the model reached 0.886 / 0.586. With two it reached 0.890 / 0.601, which is statistically indistinguishable from the reference's four layers. Therefore, the convolutions matter, as adding the first conv layer is the largest single improvement. However, beyond one or two layers we see no gain, and each extra layer costs training time.

Part C: How long should the window be?

With a 0.67 s window (20 samples), the model reached 0.899 weighted F1 and 0.636 macro F1. These were the highest scores of any run, though from a single seed. With a 2 s window (60 samples), it reached 0.886 and 0.602, no better than the reference, and each epoch took about four times longer to train. The short window helped the drawer gestures most, where their average F1 was 0.566, compared with 0.419 for the reference and 0.443 for the 2 s window. Therefore, a shorter window works better, mainly because it helps the short drawer gestures. This is consistent with the paper's finding that short gestures benefit from being fully contained in the window. A longer window brings no major improvements and is much slower to train.

Part D: How much regularization is needed?

The fully regularized model (all three active) scored 0.886 weighted / 0.578 macro F1. Without dropout it scored 0.888 / 0.573, without weight decay 0.893 / 0.613, and without augmentation 0.887 / 0.553. Regularization did change the validation loss. With all three regularizers, validation loss stayed low, reaching its minimum at epoch 68, but for the reference it started rising after about 10 epochs. However, this did not translate into better test F1. Therefore, the model needs very little regularization, as even removing dropout made no measurable difference.

Part E: Does a learning-rate warm-up make training more stable?

Training with warm-up started more slowly, where validation weighted F1 after the first epoch was 0.735, against about 0.80 for the reference, and it reached 0.87 at epoch 10 (within the reference's range of 8-18). During the first 15 epochs its largest drop in validation F1 from one epoch to the next was 0.026, slightly larger than in any reference run. On the test set, it scored 0.894 weighted F1 and 0.580 macro F1, which is not a clear change from the reference. Therefore, warm-up did not make training more stable. At the paper's learning rate, RMSProp already trains smoothly, so there is no early instability for the warm-up to improve.

Part F: Does handling class imbalance help rare gestures?

After implementing focal loss, weighted F1 stayed the same as the reference, but macro F1 rose from 0.576 to 0.619, about twice the reference's seed-to-seed variation. The drawer gestures improved most, where their average F1 rose to 0.545. None of them stayed below 0.40, in comparison to each reference run that had two to four drawer gestures below that level. Recall improved most consistently, where on average the gestures were detected more often than in any reference run. Additionally, recognition of Null was almost unchanged, and not every gesture improved. Therefore, handling the imbalance does help the rare gestures. Notably, by weighted F1, the focal loss model and the reference are identical. Only macro F1 and the per-gesture scores show the difference, which suggests the paper's weighted F1 hides how well its model recognizes the rare gestures.